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Week 6 Baseline Model Report
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Baseline Model Report

Link to GitHub notebooks: Model Evaluation

Link to 1 minute video: 1-Minute Clinical Explainer Video

Objective

The objective was to build and evaluate two baseline machine-learning models for predicting Emergency Severity Index (ESI) triage levels (1–5) using information available at triage. A logistic regression and a decision tree were compared against a stratified dummy classifier to determine whether they learned useful patterns rather than just guessing, with particular attention to whether they correctly identified ESI Level 1, the most critical and least frequent of the five classes.

Dataset and feature selection

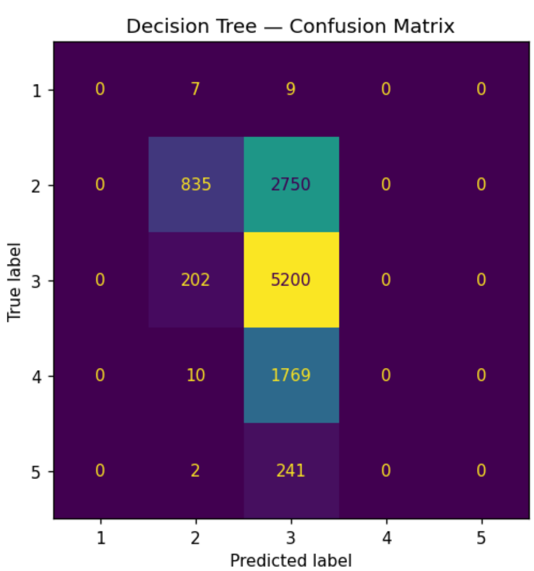
The dataset contains 55,121 emergency department visits and 225 variables, split 80/20 into training and testing sets (stratified, random_state = 42) so each ESI level is represented in similar proportions in both sets.

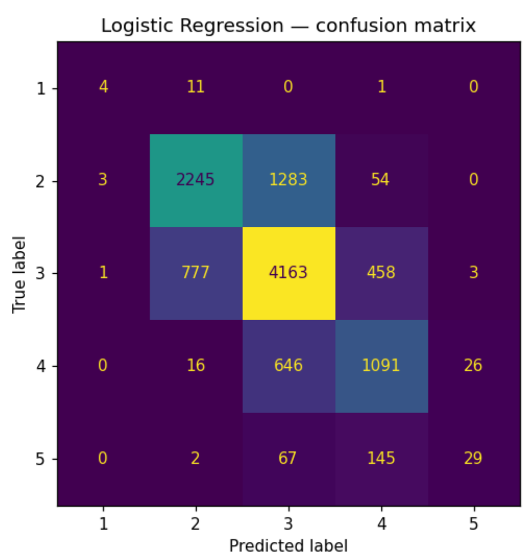
Only numeric features available at the time of triage were used (age, triage vital signs and numeric chief-complaint indicators). The ESI target and any post-triage outcomes were excluded to prevent data leakage and ensure that the models used only information available when the triage decision was made. Non-numeric variables, such as race, ethnicity, gender, insurance status and arrival information, were also excluded.

Baseline models

The dummy classifier provided a simple comparison by making predictions based on how common each ESI level was in the dataset. Logistic regression was trained after putting the numeric features on a similar scale. A decision tree was also tested using a depth of 5 so that it would remain simple enough to interpret and be less likely to fit the training data too closely.

Results





Model	Accuracy	Macro F1	Weighted F1	ESI 1 Recall
Dummy Classifier	0.375	0.204	0.375	0.00
Logistic Regression	0.683	0.508	0.677	0.25
Decision Tree	0.547	0.207	0.449	0.00

Logistic regression performed best overall, correctly predicting about 68% of cases compared with about 55% for the decision tree and 38% for the dummy model. However, its performance was still stronger for the common middle categories than for the rarer levels at either end of the urgency scale. Macro F1 gives equal importance to performance across all five ESI levels, while weighted F1 weights each level according to the number of patients in it. This is why weighted F1 is higher even when the model performs poorly on the rare classes.

Performance Across ESI Levels

Accuracy and weighted F1 are influenced more by the common middle classes, so the confusion matrices provide a clearer idea of performance at each level.

ESI Level	n (test set)	Logistic Regression Recall	Decision Tree Recall
1 (most critical)	16	0.25	0.00
2	3,585	0.63	0.23
3	5,402	0.77	0.96
4	1,779	0.61	0.00
5 (least urgent)	243	0.12	0.00

Logistic regression performs reasonably on ESI Levels 2, 3 and 4, correctly identifying between 61% and 77% of these cases. However, it struggled with the rarest levels. It correctly identified only 25% of ESI Level 1 patients and 12% of ESI Level 5 patients.

The decision tree performed well mainly on ESI Level 3, correctly identifying 96% of those cases. It identified only 23% of Level 2 patients and none of the Level 1, 4 or 5 patients. This shows that it relied heavily on the common middle categories rather than learning to separate all five levels.

Primary metric: Recall for ESI Level 1

Recall for ESI Level 1 was chosen as the main measure because it shows how many critical patients the model correctly identifies. ESI 1 cases are uncommon in this dataset, as they often are in emergency department settings, so overall accuracy can be misleading. A model could appear strong overall while still missing patients who require immediate treatment. Identifying as many true ESI 1 cases as possible is therefore more clinically important than maximising overall accuracy.

Failure-mode reflection

The most concerning failure was the under-identification of ESI Level 1 patients. Logistic regression correctly identified 4 of the 16 true ESI 1 cases and missed the remaining 12, giving it a recall of 0.25. The decision tree and dummy baseline did not identify any ESI 1 cases. In clinical use, these errors could assign critically ill patients a lower triage priority and delay assessment or treatment.

A closer review showed that stroke alert was the most common chief complaint among the ESI Level 1 patients and accounted for all four cases correctly identified by logistic regression. The 12 missed cases had a wider range of complaints, including shortness of breath, altered mental status, hypotension, poisoning and respiratory distress. Their median vital signs were also generally similar to the other ESI groups, which may have made them harder for the model to recognise. However, these patterns should be interpreted carefully because the test set contained only 16 ESI Level 1 patients.

These results are based on one train/test split and have not yet been tested across different splits, patient groups or a Caribbean emergency department.

Next Steps

Future work will focus on improving detection of ESI Levels 1 and 5, since both models performed poorly at the extremes. Planned steps include:

- Testing class-weighted models or resampling to improve recall for ESI Levels 1 and 5.
- Reviewing chief complaints, vital signs, and incorrect predictions to identify which features best distinguish ESI Levels 1 and 5.
- Testing more advanced classifiers, such as random forest or gradient boosting.
- Repeating the evaluation across multiple train/test splits to check whether the results are stable.
- Comparing models using a wider range of performance measures beyond overall accuracy.

Conclusion

This baseline work shows that information collected during triage can help predict ESI levels, and logistic regression performed better than the decision tree on every measure. However, both models performed reasonably well in the middle of the distribution and struggled at its edges. Logistic regression identified only 1 in 4 of the most critical patients and about 1 in 8 of the least urgent patients, while the decision tree relied almost entirely on the middle categories and missed nearly every patient at either extreme. Improving performance across the full range of triage levels is therefore necessary before either model could be considered for clinical use.